

KANSAI INTL, Japan

WMO: 477740

Lat: 34.434N

Lon: 135.233E

Elev: 8

StdP: 101.23

Time Zone: 9.00 (E09)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	2.8	-8.0	1.9	3.6	-6.9	2.1	4.3	14.4	5.2	13.2	5.0	6.9	290	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	33.0	25.5	32.1	25.3	31.1	25.2	26.8	30.2	26.5	29.8	26.1	29.3	5.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.4	28.8	25.2	20.4	28.5	25.0	20.1	28.4	84.7	30.6	82.6	29.9	80.7	29.6	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.5	10.9	9.5	-0.1	34.7	1.5	0.9	-1.2	35.4	-2.1	35.9	-2.9	36.4	-4.0	37.0		
			-3.0	27.6	1.2	0.8	-3.9	28.2	-4.6	28.6	-5.3	29.1	-6.2	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	6.6	6.7	9.6	14.5	19.4	23.1	27.2	28.8	25.4	20.1	14.5	9.3
	DBStd	8.02	2.02	2.34	2.64	2.77	2.12	1.94	2.04	1.58	2.44	2.52	2.77	2.53
	HDD10.0	290	108	96	40	2	0	0	0	0	0	0	1	43
	HDD18.3	1499	365	326	270	117	13	0	0	0	0	11	118	279
	CDD10.0	2903	1	4	28	137	292	393	533	584	462	312	135	22
	CDD18.3	1072	0	0	0	2	47	143	275	326	212	65	3	0
	CDH23.3	8894	0	0	0	3	90	589	2664	3759	1631	157	1	0
	CDH26.7	2797	0	0	0	0	4	80	834	1482	388	10	0	0
Wind	WSAvg	4.4	5.3	4.7	4.6	4.2	3.8	3.7	4.0	4.1	4.2	4.4	4.6	5.3
Precipitation	PrecAvg	1468	50	70	103	107	154	196	163	114	199	145	105	63
	PrecMax	1925	125	167	188	229	438	382	409	469	428	336	313	153
	PrecMin	750	5	22	55	38	44	78	42	10	36	32	18	5
	PrecStd	285	31	37	34	44	92	77	97	95	102	78	65	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	16.0	18.2	23.2	27.1	30.2	34.0	34.2	32.2	28.0	22.2	17.8
		MCWB	9.9	13.6	13.6	16.9	18.8	23.8	25.9	25.6	25.2	22.8	18.0	14.8
	2%	DB	11.9	13.1	16.9	21.2	25.2	28.8	32.8	33.2	31.1	26.1	21.0	15.2
		MCWB	8.7	9.9	13.2	16.0	18.6	23.1	25.7	25.5	24.7	21.1	17.3	11.9
	5%	DB	10.9	11.8	15.2	20.0	24.1	27.2	31.8	32.8	30.0	25.0	20.0	14.1
		MCWB	7.7	8.8	11.5	15.4	18.0	21.8	25.5	25.4	23.9	20.4	16.2	10.7
	10%	DB	9.9	10.2	14.0	18.9	23.0	26.2	30.2	31.9	28.9	24.0	19.0	13.1
		MCWB	6.8	7.4	10.6	14.7	17.6	21.6	25.1	25.3	23.5	19.1	15.3	9.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	13.8	15.4	19.4	22.1	25.3	27.4	27.7	26.7	24.3	19.6	15.5
		MCDB	13.0	15.4	17.1	21.2	24.2	28.4	30.6	30.4	29.9	26.4	20.9	17.0
	2%	WB	9.6	11.0	13.6	17.7	20.7	24.5	26.7	27.1	26.1	22.5	18.2	12.9
		MCDB	10.7	12.7	15.9	19.7	23.0	27.1	30.2	30.4	28.9	25.1	20.2	14.9
	5%	WB	8.2	9.5	12.4	16.4	19.7	23.8	26.3	26.6	25.4	21.3	17.0	11.4
		MCDB	10.1	10.9	14.7	19.0	22.3	26.0	29.6	30.0	28.3	23.9	19.2	13.6
	10%	WB	7.0	8.1	11.1	15.3	18.7	22.9	25.8	26.3	24.7	20.2	15.7	10.3
		MCDB	9.2	9.9	13.6	18.0	21.7	25.2	29.0	29.6	27.7	23.0	18.3	12.7
Mean Daily Temperature Range	5% DB	MDBR	4.4	5.0	6.0	6.7	6.8	5.7	5.5	5.8	5.3	5.2	4.9	4.5
		MCDBR	5.6	6.8	7.7	8.1	8.2	7.0	6.7	6.7	6.2	6.0	5.8	5.7
		MCWBR	4.8	5.5	5.7	4.7	4.1	3.0	2.6	2.6	3.0	3.8	4.9	5.0
	5% WB	MCDBR	5.3	6.2	7.3	6.9	6.6	5.5	6.2	5.9	5.4	5.3	5.6	5.5
		MCWBR	5.2	5.8	6.3	5.0	4.5	3.1	2.7	2.7	3.1	4.3	5.4	5.5
Clear-Sky Solar Irradiance	taub	0.403	0.448	0.509	0.543	0.547	0.566	0.568	0.536	0.494	0.454	0.423	0.398	
	taud	2.157	2.050	1.904	1.876	1.933	1.963	2.004	2.083	2.172	2.199	2.187	2.183	
	Ebn at Noon	774	779	763	760	762	746	741	758	771	767	748	752	
	Edh at Noon	119	147	185	199	190	184	176	160	140	126	115	110	
All-Sky Solar Radiation	RadAvg	2.38	3.01	3.98	4.81	5.23	4.69	5.06	5.19	4.07	3.30	2.54	2.15	
	RadStd	0.17	0.22	0.30	0.43	0.53	0.45	0.63	0.45	0.35	0.33	0.21	0.13	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page